

MUHAMMAD RAIHAN RIZKY ZAIN

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Summary

Computer Science Undergraduate student at Universitas Padjadjaran with an interest in building useful software solutions and currently exploring how AI can also make them more intelligent and impactful. Through projects and continuous learning, I enjoy turning ideas into practical digital products by understanding user needs, designing solutions, and continuously improving them through experimentation and iteration. I’m particularly interested in creating software that effectively solves problems using thoughtful and reliable technical solutions. My current interests include Software Development and Artificial Intelligence, and I’m always excited to explore new technologies and collaborate with others to grow, exchange ideas, and build meaningful solutions.

Education

Universitas Padjadjaran <i>Bachelor’s Degree in Computer Science, GPA: 3.86/4.00</i>	Jatinangor, West Java, Indonesia <i>Aug 2023 – Aug 2027 (Expected)</i>
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Experience

Backend Engineer Intern <i>Universitas Padjadjaran (Teaching and Learning Innovation)</i>	Feb 2026 - Present <i>Jatinangor, West Java, Indonesia</i>
- Engineered the core RESTful API for a Question Bank Management System using Laravel 11 and optimized PostgreSQL database schemas. - Developed advanced features (autosave, bulk actions, Word/Excel import/export) and integrated Redis caching for low-latency data retrieval. - Containerized the application stack using Docker Compose and documented API contracts via Swagger/OpenAPI to streamline testing and integration.	
AI Engineer Cohort <i>Coding Camp powered by DBS Foundation - Indonesia</i>	Feb 2026 – Present <i>[Remote]</i>
- Selected as an AI Engineer Cohort participant in Coding Camp powered by DBS Foundation, with a focus on machine learning, deep learning, and AI system integration. - Contributed to the final capstone project, “CuanSelor,” a fintech-based financial advisor application, by developing AI-related components including financial profile prediction, exploring LLM-powered financial advisory features, and supporting model integration through API deployment. - Collaborated with cross-functional team members across Data Science and Full-Stack Development to deliver an integrated solution.	
Artificial Intelligence Laboratory Assistant <i>Informatika Universitas Padjadjaran</i>	Feb 2026 – Present <i>Jatinangor, West Java, Indonesia</i>
- Assisted students in understanding Artificial Intelligence concepts through practical, hands-on sessions. - Managed teaching activities, including assignments, assessments, and grading.	
Software Developer <i>medphoto.booth</i>	Oct 2025 – Present <i>[Remote]</i>
- Co-developed server-side infrastructure and a specialized Node.js server to manage real-time data ingestion and multi-laptop synchronization across remote capture stations. - Engineered automated media pipelines using Sharp and Omggif for real-time image enhancement, custom GIF generation, and local asset synchronization. - Integrated a robust email delivery system via Nodemailer with SMTP fallbacks to automate digital media distribution to end-users.	
IoT Engineer <i>Samsung Innovation Campus (Batch 6)</i>	Feb 2025 – May 2025 <i>[Remote]</i>
- Engineered a scalable data pipeline using ESP32 microcontrollers to capture and transmit real-time sensor data via MQTT/HTTP. - Designed a MongoDB database for telemetry logs and built a Flask API with a Streamlit dashboard for real-time monitoring. - Achieved Top 1 Ranking during Stage 2 evaluation and advanced to Stage 3 (Top 500 national innovators).	

Organizational Experience

Vice Head of Information Media Bureau <i>HIMATIF FMIPA Universitas Padjadjaran</i>	Jan 2025 – Dec 2025 <i>Jatinangor, Indonesia</i>
• Led branding and digital communication strategies for the Informatics Student Association, ensuring a cohesive public identity. • Managed multi-platform digital content workflows, web assets, and public relations materials.	
Head of Publication, Design & Documentation <i>Informatics Fun Day 2024</i>	Oct 2024 – Nov 2024 <i>Jatinangor, Indonesia</i>
• Managed and directed a creative team of 10 members handling event promotion, visual asset production, and multimedia storytelling. • Designed comprehensive visual identities, marketing assets, merchandise, and event branding layouts. • Orchestrated strict publication timelines, ensuring high-quality creative deliverables across all digital channels.	

Selected Projects

LearnMate AI AI Engineer
• Developed an intelligent, Retrieval-Augmented Generation (RAG) educational chatbot designed to ground LLM responses strictly on specific lecture materials. • Fine-tuned semantic retrieval models using a custom triplet dataset to maximize document relevance and minimize AI hallucinations. • Engineered a robust, two-stage retrieval pipeline combining fast vector search with an advanced reranking mechanism.
NutriGrow AI & Backend Engineer
• Developed an AI-powered stunting classification system aligned with SDG 2.2 to support targeted child growth monitoring and health assessments. • Engineered core backend services and integrated workflows for tele-nutrition consultations, maternal/adolescent health logs, and product recommendation pathways. • Conducted external usability testing with mothers and teenagers to evaluate and optimize the platform’s real-world clarity, trust, and user experience.
PasarAjaibAR Unity AR Engineer
• Developed an interactive, educational Mobile Augmented Reality game designed for 2nd and 3rd-grade elementary students to address SDG 4.1 (Quality Education). • Implemented precise surface tracking and spatial integration using plane detection mechanisms powered by ARCore and AR Foundation. • Designed immersive 3D object interactions and child-friendly touch controls to deliver an engaging, hands-on spatial learning experience.
Practicum Attendance System Java Full-Stack Developer
• Developed a desktop-based management system using Java NetBeans, implementing the MVC (Model-View-Controller) pattern to separate data logic from the user interface. • Designed and integrated a relational database using MySQL, handling relationships between students, teaching assistants, and course schedules. • Engineered core functionalities including secure user authentication, real-time attendance tracking, and automated scheduling for laboratory sessions. • Designed an interactive Graphical User Interface (GUI) to facilitate seamless data management for administrators and students.

Skills, Certifications & Additional Experience

Artificial Intelligence & Data: Python, PyTorch, Scikit-Learn, Sentence Transformers, FAISS, TensorFlow, LangChain, Machine Learning, OpenCV, Image Processing
Software Development: PHP (Laravel), JavaScript, TypeScript, Java, Node.js, Express.js, Next.js, Flask, Streamlit, C#, Unity
Database & DevOps: PostgreSQL, MongoDB, MySQL, Redis, Linux (Fedora/Arch), Docker, Tailscale, Nginx
Tools & Version Control: Git, GitHub, VS Code, Figma, Swagger, Postman, Notion
Design & Multimedia: Adobe Illustrator, Adobe After Effects, DaVinci Resolve
Languages: Bahasa Indonesia (Native), English (Upper-Intermediate / Professional Fluency)